

A disproof of Conjecture 00000000314

Every $M(\sqrt{p})$ is at least $2\sqrt{2}$, so $\{M(\sqrt{p})\}$ is not dense in $[\sqrt{5}, 3)$

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1 The conjecture

Conjecture 00000000314 reads (English text, verbatim): “Definition: The Markov constant $M(\alpha) = \limsup 1/\|q\alpha\|$ measures the quality of worst approximation. Conjecture: The set $\{M(\sqrt{p}) : p \text{ prime}\}$ is dense in the half-open interval from $\sqrt{5}$ to 3; and the density follows from the transition statistics of the continued-fraction partial quotients of square roots of primes.” The Chinese text gives the same definition and the same interval.

The statement is a conjunction. We refute its first conjunct (density); the second conjunct (“the density follows from ...”) presupposes the first and falls with it.

2 Definitions and readings

Definition 1. For $x \in \mathbb{R}$, $\|x\| = \inf_{n \in \mathbb{Z}} |x - n|$ is the distance from x to the nearest integer (Lean: `distNearestInt`). Limits superior are taken over positive integers $q \rightarrow \infty$ and values lie in $[0, \infty]$, with the convention $1/0 = \infty$.

We treat two readings of M .

- **Standard reading** (Lean: `markovConst`): $M(\alpha) = \limsup_{q \rightarrow \infty} \frac{1}{q \|q\alpha\|}$ ($= 1/\liminf_q q\|q\alpha\|$). This is the usual Lagrange value of α ; its values on irrationals form the Lagrange spectrum, whose smallest element is $\sqrt{5}$, which is why the text’s interval starts at $\sqrt{5}$.
- **Literal reading** (Lean: `markovConstLiteral`): $M(\alpha) = \limsup_{q \rightarrow \infty} \frac{1}{\|q\alpha\|}$, exactly as printed (no factor q).

“ S is dense in $I = [\sqrt{5}, 3)$ ” is formalized in two ways, and both are refuted: (T) $I \subseteq \overline{S}$ (Lean: `DenseInTop`); (I) for all $a < b$ in I there is $s \in S$ with $a < s < b$ (Lean: `DenseInIntervals`). Both are taken in $[0, \infty]$ (Lean type `ENNReal`), and $S = \{M(\sqrt{p}) : p \text{ prime}\}$ (Lean: `valueSet`).

3 The proof

Lemma 2 (Pell estimate; Lean: `pell_bound`). *Let $d \geq 1$ and let x, y be positive integers with $x^2 - dy^2 = 1$. Then $\|y\sqrt{d}\| \cdot 2y\sqrt{d} \leq 1$.*

Proof. Put $s = \sqrt{d}$. Since $(x - ys)(x + ys) = x^2 - dy^2 = 1$ and $x + ys > 0$, we get $x - ys > 0$, so $x > ys$. Hence $\|ys\| \leq |ys - x| = x - ys$, and $\|ys\| \cdot 2ys \leq (x - ys) \cdot 2ys \leq (x - ys)(x + ys) = 1$. $\square$

Lemma 3 (Lean: `pell_large, exists_good_q`). *For every prime p and every N there is an integer $q > N$ with $\|q\sqrt{p}\| \cdot 2q\sqrt{p} \leq 1$.*

Proof. A prime is not a perfect square, so Pell’s equation $x^2 - py^2 = 1$ has a fundamental solution a (Mathlib: `Pell.IsFundamental.exists_of_not_isSquare`). The map $n \mapsto y(a^n)$ is strictly increasing on $\mathbb{Z}$ (Mathlib: `Pell.IsFundamental.y_strictMono`) with $y(a^0) = 0$, so $y(a^n) \geq n$ for $n \in \mathbb{N}$; and $x(a^n) > 0$ (Mathlib: `Pell.Solution_1.x_zpow_pos`; in Mathlib the subscript 1 is a Unicode subscript). Take $q = y(a^{N+1})$ and apply Lemma 1. $\square$

Theorem 4 (Lean: `markovConst_sqrt_prime_ge, markovConstLiteral_sqrt_prime`). *For every prime p : (a) $M(\sqrt{p}) \geq 2\sqrt{p}$ in the standard reading; (b) $M(\sqrt{p}) = \infty$ in the literal reading.*

Proof. (a) For the q of Lemma 2, $q\|q\sqrt{p}\| \cdot 2\sqrt{p} \leq 1$, i.e. $1/(q\|q\sqrt{p}\|) \geq 2\sqrt{p}$. Such q exist beyond every N , so the limsup is at least $2\sqrt{p}$. (b) For every real B , apply Lemma 2 with $N \geq B$: then $1/\|q\sqrt{p}\| \geq 2q\sqrt{p} \geq q \geq B$, because $\sqrt{p} \geq 1$. Thus the limsup is at least every real B , so it is ∞ . $\square$

Theorem 5 (Lean: `not_dense_of_ge, conjecture314_false`). *In both readings of M , the set $S = \{M(\sqrt{p}) : p \text{ prime}\}$ is dense in $[\sqrt{5}, 3)$ in neither sense (T) nor sense (I). In fact $S \cap (\sqrt{5}, 2\sqrt{2}) = \emptyset$ while $\sqrt{5} < 2\sqrt{2} < 3$.*

Proof. By Theorem 3, every element of S is $\geq 2\sqrt{p} \geq 2\sqrt{2}$ (standard) or $= \infty$ (literal), so $S \subseteq [2\sqrt{2}, \infty]$, a closed set. Also $\sqrt{5} < 2\sqrt{2} = \sqrt{8} < \sqrt{9} = 3$. (T): $\sqrt{5} \in [\sqrt{5}, 3)$ but $\sqrt{5} \notin \overline{S} \subseteq [2\sqrt{2}, \infty]$. (I): take $a = \sqrt{5}$, $b = 2\sqrt{2}$, both in $[\sqrt{5}, 3)$; no $s \in S$ satisfies $s < b$. $\square$

Remark 6. For context only (not used in the proof): Wikipedia’s article on the Markov spectrum states that “the initial part of the Markov and Lagrange spectrum lying in the interval $[\sqrt{5}, 3)$ are both equal and they are a discrete set” (https://en.wikipedia.org/wiki/Markov_spectrum). The argument above is self-contained and gives the stronger statement that no $M(\sqrt{p})$ lies below $2\sqrt{2}$.

4 Lean formalization

File `lean/Conjecture314/Basic.lean` (Lean 4, Mathlib v4.33.1, only import `Mathlib`). Main theorem:

```
theorem conjecture314_false :
  (Not (DenseInTop (valueSet markovConst) targetInterval) /\
   Not (DenseInIntervals (valueSet markovConst) targetInterval)) /\
  (Not (DenseInTop (valueSet markovConstLiteral) targetInterval) /\
   Not (DenseInIntervals (valueSet markovConstLiteral) targetInterval))
```

(ASCII rendering: the Lean source writes the negations with the Unicode symbol for Not and the conjunctions with the Unicode wedge.) Here `targetInterval = Set.Ico (ENNReal.ofReal (sqrt 5)) 3`. The full chain of lemmas: `distNearestInt_le, pell_large, pell_bound, exists_good_q, markovConst_sqrt_prime_ge, markovConstLiteral_sqrt_prime, sqrt5_lt_two_sqrt2, two_sqrt2_lt_three, not_dense_of_ge`.

Axiom `audit. #print axioms C314.conjecture314_false` prints `[propext, Classical.choice, Quot.sound]`. There is no `sorry` and no `native_decide`.

Conventions. q ranges over $\mathbb{N}$ with $\limsup$ along **atTop** (the term $q = 0$ is irrelevant to a $\limsup$); values are in $[0, \infty]$ with $1/0 = \infty$; $\sqrt{\cdot}$ is the nonnegative square root. Using $q \in \mathbb{Z}$, $|q| \rightarrow \infty$ gives the same values, since $\|-x\| = \|x\|$.